

Design and Experimental Validation of a High-Power-Density GaN-Based Inverter Module for Integrated Modular Motor Drives in All-Electric Aircraft Applications

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Abstract—The integrated modular motor drive (IMMD) concept is a promising solution for transportation electrification. IMMDs offer essential features such as high power density, efficiency, and reliability which are critical for electrified propulsion systems. While integration provides advantages in power density and efficiency, it also presents significant design challenges due to space constraints. In this study, a single inverter module of a 250 kW IMMD is designed and tested, which functions as an autonomous power converter. Each module contains a power converter, dc-link capacitor bank, and a local microcontroller, enabling independent control. The modules of the targeted IMMD are identical, allowing for easier replacement of failed units and reducing production complexity and cost. The designed inverter module underwent testing to evaluate both its switching characteristics and continuous operation. The manufactured inverter module obtained a gravimetric power density (specific power) of 43.96 kW/kg and an efficiency of 98.65%.

Index Terms—All-electric aircraft, electrothermal design, finite element analysis (FEA), gallium nitride (GaN), integrated modular motor drive (IMMD), PCB busbar.

I. INTRODUCTION

TRANSPORTATION accounts for almost 30% of carbon emissions, making the electrification of transportation a central focus of research and development in both industry and academia over the past decades [1]. The global market share of electric vehicles (EVs) has increased to 18% as of 2023, highlighting the rapid transition to electrified transportation [2]. To support this transition, fundamental advancements in electric machine and drive system technologies are necessary. Programs like CIRCUITS and ASCEND, introduced by ARPA-E, a U.S. government agency under the DOE, aim to develop the next generation of power converters with high power density and efficiency [3], [4].

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Compared to bulky and heavy industrial drive systems, drive units for electrified transportation must be more efficient, lightweight, and reliable [5]. Traditionally, drive units were connected to electric motors via long and bulky cables, leading to reduced efficiency and insulation concerns due to the reflected wave phenomenon [6]. To meet the high efficiency and power density requirements for transportation-level propulsion systems, the traditional structure of the machine and drive system must be altered. The integrated motor drive (IMD) concept addresses this by integrating the electric motor and drive system into a single structure [7]. This integration significantly improves overall system cost, efficiency, and power density [8]. The concept can be further enhanced by modularizing the system, distributing the rated torque/power demand among identical modules, which increases reliability and fault tolerance, giving rise to the concept of integrated modular motor drive (IMMD) [9].

From the drive system standpoint, power density and efficiency are highly dependent on semiconductor selection. While silicon (Si) semiconductors reached their material theoretical limits, the emergence of wide bandgap (WBG) semiconductors paved the way for further improvements in the drive system domain [10]. WBG devices, such as GaN and SiC, exhibit excellent characteristics, including high junction temperature limit, low on-state resistance, and low reverse recovery losses, making them ideal for IMMD applications [11]. Consequently, ongoing research focuses on developing IMDs and IMMDs using GaN or SiC semiconductors.

A subset of the studies in the literature is dedicated to the design of the IMMDs where the development of electric machines and drive units is discussed. For instance, in [12], the design of a five-phase, 110 kW SiC-based IMMD is discussed. The authors claim that with the integration of the motor and drive unit, the loss density of the motor is increased significantly which leads to a temperature increase in the overall system. Therefore, the thermal management system (TMS) design is important for the system's reliable functioning and achieving the aggressive target for power density. Moreover, in their motor design section, they used the skewed rotor method to decrease the torque ripple. In [13], a low-power GaN-based IMMD is developed for a drone

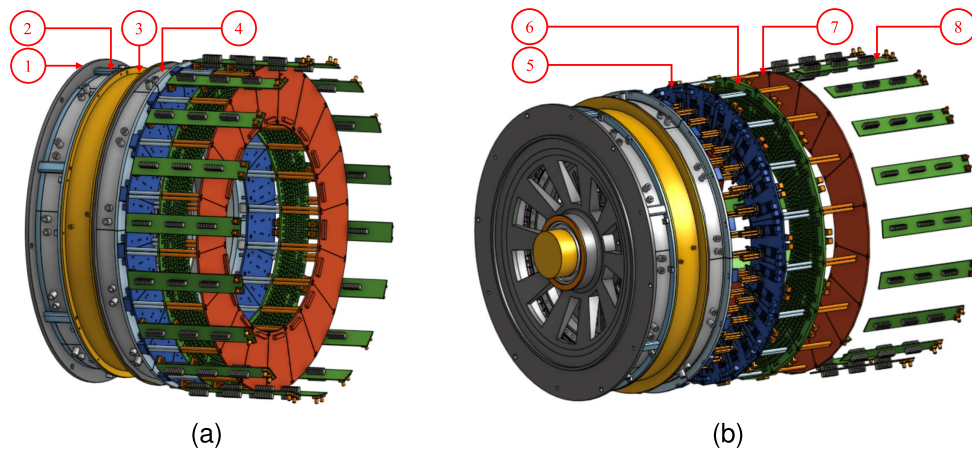


Fig. 1. Overall electric machine and IMMD representation. (a) 3-D view of the overall system: 1) housing; 2) dedicated condenser for motor coils; 3) stator; and 4) shared condenser between motor coils and IMMD. (b) Exploded view of the IMMD: 5) GaN-based FB converter PCB (blue PCB); 6) dc-link capacitor PCB (green PCB); 7) communication PCB (red PCB); and 8) interconnecting dc busbar.

application. The high-power density target is claimed as the reason for GaN utilization where Infineon CoolGaN_s with low thermal resistance packaging were used. It was claimed that the interconnecting cables and connectors between the inverter and the motor are failure sources that are eliminated through the integration of the drive and the motor. The obtained power density of the drive unit is 9.47 kW/kg. Wang et al. [14] developed a GaN-based IMMD for an induction motor. They connected the GaN-based inverter modules in series with the dc supply, enabling the use of lower voltage GaN devices. Also, by interleaving the gate signals the rms current requirement of dc link was reduced leading to dc-link capacitor size reduction, and consequently an increase in the power density. It can be inferred that, a part of the design-related studies contains the dc-link capacitor selection, sizing, current ripple reduction strategies, and busbar design [15], [16].

A comprehensive characterization of conducted EMI in a 211 kW SiC-based inverter designed for aviation propulsion systems is presented in [17]. The study introduces effective EMI mitigation techniques, including dc and ac filter designs, optimized PCB layout to minimize loop inductance, and grounding strategies for reliable operation. The proposed design was tested where an efficiency of 99% and a power density of 19.5 kW/kg were reported [18].

Research also covers modeling and control of modular motors and IMMDs. Unlike conventional motors with only three exposed winding ends, modular motor structures have both coil ends accessible, necessitating different motor models based on the coil connections and mutual interactions [19]. A multistator (MS)-based modular vector control scheme is proposed in [20], decoupling the machine into independent three-phase subsets, each controlled by dedicated drive units. Having the overall system model, the next step is to design the control system. Considering the modular aspect of the IMMDs, a part of the control system design focus is on distributing the rated demand torque/power among the modules [21], [22]. This, in turn, emboldens the fault-tolerant capability and reliability of the IMMDs. Verkroost et al. [23] used a deterministic static average consensus algorithm which is inspired by the approach in economic dispatch in a smart grid.

In other words, the overall torque demand is distributed among different modules in a decentralized manner.

One of the key challenges in the development of a tightly integrated system is the design of the TMS [24]. The overall system power density is highly influenced by the TMS for both the motor and drive unit. In addition, an optimized shared TMS for both components is advantageous in terms of cost-effectiveness [25]. Torres et al. [26] studied various cooling strategies for a 3 kW IMMD with different enclosures. Their results showed that despite the close proximity of the motor and drive unit (two main heat generation sources), several TMS configurations successfully controlled the temperature, suggesting promising commercialization potential for IMMDs.

In this article, the design of the inverter module for a 250 kW IMMD aimed at all-electric aircraft applications is presented. The overall system is depicted in Fig. 1, illustrating the assembly of the IMMD and its integration with the motor. This novel modular topology features individual inverter modules, each equipped with a dedicated microcontroller board and a dc-link capacitor bank, effectively making each module a fully autonomous power converter. The system modularity and individual module autonomy offers significant advantages, including fault tolerance and simplified maintenance, as only the affected modules need replacement in the event of a failure. Furthermore, the selected topology minimizes current ripple, which consequently reduces torque ripple—an essential factor in transportation applications where reliability, smooth operation, and passenger comfort are paramount [27]. While gravimetric power density (specific power), which is proportional to weight, is a critical constraint in aviation applications, volumetric power density is not a limiting factor in this application. To address the space limitations, a practical solution of stacking the PCBs in the *z*-direction is implemented.

The paper's contributions are as follows.

- 1) An in-depth discussion rich in fundamental engineering, design processes, decision-making, and insight is provided following the design of a fully integrated inverter module, which serves as a building block of an IMMD system in an all-electric aviation application.

The discussion encompasses the following key aspects: a) selection of the power converter topology to supply a single motor coil; b) definition of criteria for power device selection and detailed selection process; c) electrothermal simulations of the selected topology with the chosen power device to determine the number of parallel devices and TMS requirements; d) detailed design of the power converter PCB, addressing challenges such as power loop inductance minimization; and e) design of the dc-link capacitor bank and busbar with a focus on minimizing parasitic inductance.

- 2) Reporting of the target design metrics against experimentally verified metrics of the inverter module.

The designed inverter module's performance was validated through comprehensive experimental tests, including double pulse testing (DPT) and continuous ac testing. These tests demonstrated an efficiency of 98.65% at full load, along with a gravimetric power density of 43.96 kW/kg. In the DPT a voltage overshoot of only 3.37% was observed, verifying the success of minimizing power loop inductance. Additionally, a dv/dt of 8.42 V/ns was recorded during switching transients, which is well-suited for the target application. The article is organized as follows: in Section II, after determining the design requirements, the initial design steps are discussed, including switch selection, topology selection, and electrothermal study. In Section III, the hardware design and layout are carried out using the information obtained from Section II. Section IV demonstrates the performance of the designed system during switching transients. In Section V, the performance of the module is demonstrated during different tests. Finally, concluding remarks from the study are presented in Section VI.

II. DRIVE SYSTEM DESIGN

Integration of the drive system and the motor introduces inherent physical constraints, which must be carefully addressed during the design process. This section outlines the preliminary design steps, while adhering to these constraints. The system specifications and design targets are summarized in Table I. For aviation applications, minimizing weight is a critical priority. To achieve this, higher voltages are preferred for a given power rating, as they reduce the required current and consequently lower the weight of cables [28], [29]. However, operating at higher voltages in high-altitude environments introduces challenges such as partial discharge (PD) phenomena and insulation failure risks [30], [31]. Hence, there is a trade-off between reducing the current and subsequently the cable weight by choosing higher voltages, and protecting the insulation from PD.

In this study, the dc-link voltage, rated power, and maximum speed are predefined project requirements, while parameters such as rated coil current and frequency are derived from these requirements [32]. Additionally, there are two primary design objectives: achieving high gravimetric power density (specific power) and maximizing efficiency for the IMMD. Both objectives strongly depend on the choice of the inverter module topology and the semiconductor devices used, as well

TABLE I
SYSTEM SPECIFICATIONS AND DESIGN TARGETS

System Specifications	
Parameters	Value
V_{DC}	1 kV
P_{rated}	250 kW
ω_{max}	5000 RPM
$I_{coil,rms}$	268 Arms
$f_{o,rated}$	1.25 kHz
Design Targets	
Parameters	Value
IMMD Power Density	≥ 45 kW/kg
IMMD Efficiency (η)	$\geq 98\%$

as their associated losses. Therefore, to meet the specified targets, it is essential to select topologies and semiconductor devices that minimize losses, weight, and footprint (surface area). This careful selection process ensures that the final design satisfies the stringent requirements of the aviation application.

The integration of the drive unit with the electric motor can be implemented using various approaches, depending on their relative positioning [8]. This study adopts the axially stator-mounted (ASM) IMMD approach, where the drive modules must conform to both radial and axial constraints dictated by the motor's geometry. The target three-phase PMSM comprises 18 coil pairs, with both winding ends exposed. To meet this requirement, 18 inverter modules are employed, each dedicated to one coil pair. Fig. 2 provides a 2-D view of the relative placement of the 18 drive modules with respect to the motor's rotor and stator, emphasizing the inner radius (IRs) and outer radius (ORs) constraints imposed by the motor's design. Fig. 2 highlights the limited available surface area for each module which is calculated as follows:

$$\frac{\pi \times (OR^2 - IR^2)}{18} = 6875 \text{ mm}^2. \quad (1)$$

An inverter module, incorporating all active and passive components, control and communication hardware, and terminals, must fit within the defined surface area in (1). However, as the design is not constrained by volumetric power density, increasing the system's volume by stacking multiple PCBs with an identical surface area defined by (1) is considered a practical and effective approach.

Fig. 3 depicts a single module, representing 1/18th of the entire system illustrated in Fig. 1. Each module is composed of four key parts: the motor condenser, stator winding, shared condenser between the motor and the drive unit, and the inverter module. This study specifically focuses on the design of the inverter module, which plays a pivotal role in meeting the performance targets while complying with the constraints imposed by the integrated system architecture.

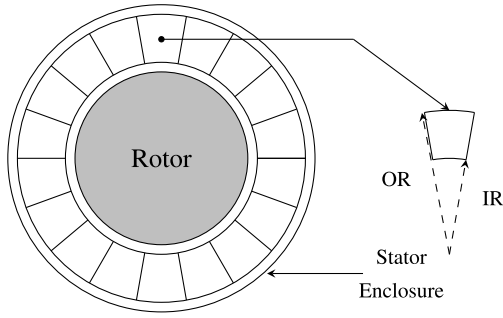


Fig. 2. Drive module geometry representation with respect to the rotor and stator (IR = 162.8 mm, OR = 256.7 mm).

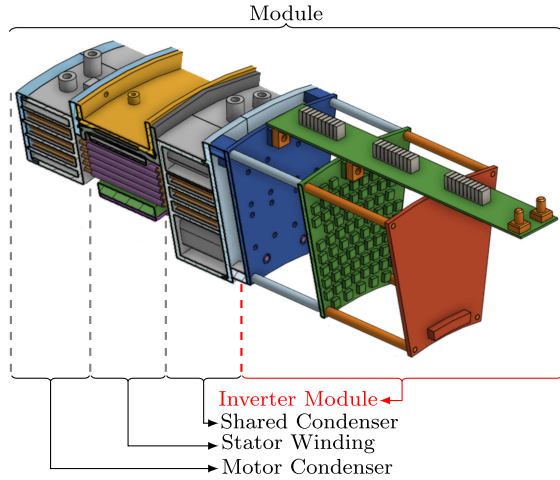


Fig. 3. 1/18th of the demonstrated system in Fig. 1 (single module).

A. Drive System Topology

Fig. 4 illustrates the overall system, showing all 18 modules and their interconnections. These modules are organized into six three-phase subsets, with three inverter modules in each subset sharing a common dc link, while the subsets are connected in series with the dc source. As depicted in Fig. 4, the circuitry of a complete module includes a motor coil pair and a drive unit (inverter module).

With limited available space as a strict constraint, a topology with the minimum number of switching positions is preferable. Since each stator coil pair can be individually controlled with both ends accessible, two topologies—half-bridge (HB) and full-bridge (FB)—are considered, as shown in Fig. 5. The HB topology reduces the number of required switching devices by half compared to the FB. However, in the HB configuration, only half of the source voltage is applied to the load due to the connection of one end of the coil pair to the midpoint of the dc-link capacitors, leading to increased load current ripple.

When selecting the topology, it is important to consider its impact on the motor performance. For instance, the PWM techniques result in induced current ripples in the motor coils which then translates to torque ripple on the motor shaft. Uncontrolled torque ripple can lead to vibrations, increased mechanical stress, and fatigue in the drivetrain, potentially compromising system integrity and performance [27]. Thus, selecting a converter with lower induced current ripple can

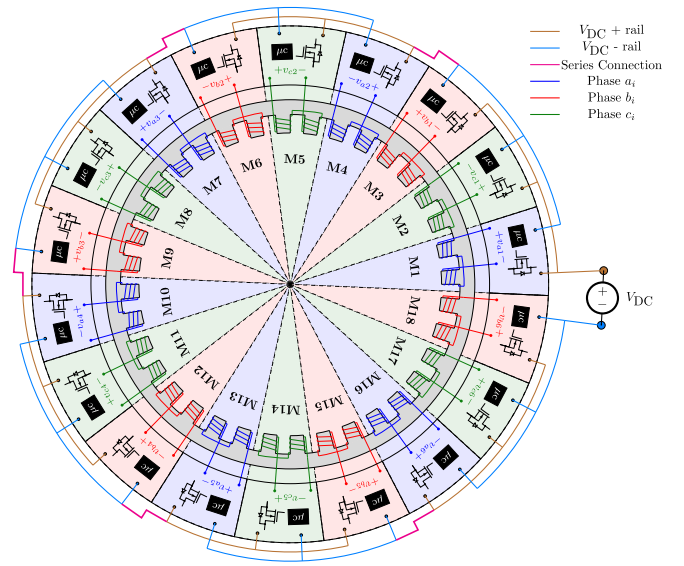


Fig. 4. Modular representation of the overall system and electrical topology of the case study IMMD. The drive unit in each module includes the power electronics converter and the control unit which is shown as μC in the figure.

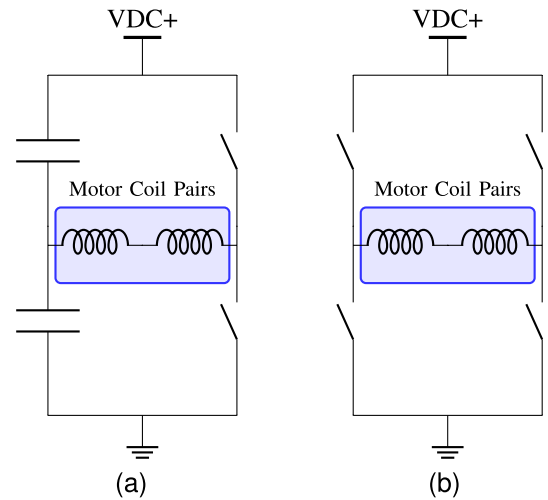


Fig. 5. Individual motor coil driver. (a) HB converter. (b) FB converter.

be considered as a torque ripple reduction strategy while designing the IMMD.

The value of maximum peak-to-peak current ripple for both HB and FB converters can be calculated through (2) and (3), respectively [33]

$$i_{\text{ripple, pp, hb, max}} = \frac{V_{dc} T_s}{2L} \quad (2)$$

$$i_{\text{ripple, pp, fb, max}} = \begin{cases} \frac{V_{dc} T_s}{4L}, & \frac{\hat{E}}{2V_{dc}} \geq 0.5 \\ \frac{\hat{E} T_s}{2L} \left(1 - \frac{\hat{E}}{2V_{dc}}\right), & \frac{\hat{E}}{2V_{dc}} < 0.5 \end{cases} \quad (3)$$

where $i_{\text{ripple, pp, hb, max}}$ and $i_{\text{ripple, pp, fb, max}}$ are the maximum peak-to-peak current ripple values for HB and FB converters, respectively, V_{dc} is the rated value of the dc-link voltage, T_s

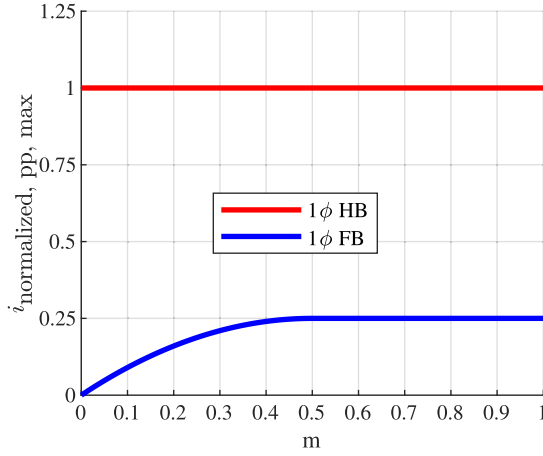


Fig. 6. Normalized maximum peak-to-peak current ripple for HB and FB topologies with respect to modulation index [33].

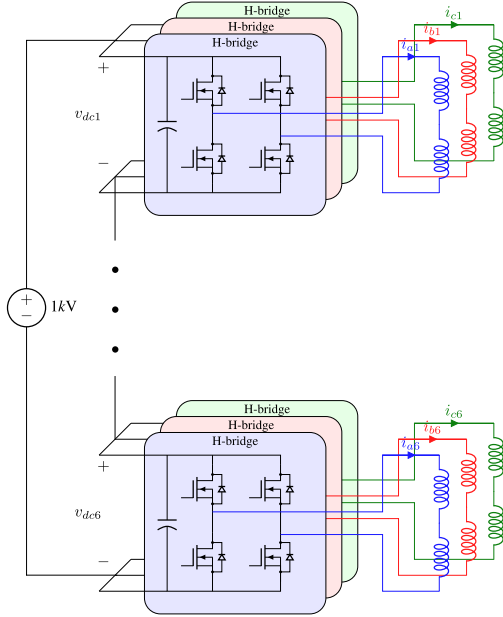


Fig. 7. Electrical topology of the designed IMMD.

is the switching period, L is the motor coil inductance, and $\hat{E}$ is the maximum value of the no-load back-EMF.

By normalizing the maximum peak-to-peak value of the current ripple (using $i_{\text{ripple, pp, hb, max}}$ as the base), the performance of the HB and FB converters can be compared in terms of the induced current ripple. Fig. 6 illustrates that the maximum peak-to-peak current ripple of an FB converter is one-fourth of an HB converter, leading to lower torque ripple, which is beneficial for motor performance.

Therefore, an FB converter is selected as the drive unit topology for this study due to its ability to reduce current/torque ripple. Fig. 7 demonstrates the electrical topology of the IMMD where each FB converter is connected to one of the motor coil pairs. In this topology, each phase contains six series of FB converters, meaning that the dc-link voltage per FB is 166.67 V.

B. Switch Selection

To satisfy the target efficiency and power density mentioned in Table I, and to accommodate the limited space for the drive system, the selected semiconductors must have low losses, small packages/footprints, and be lightweight. The ambitious design targets and unique specifications of the application, including high current demand (268 Arms), made switch selection a particularly complex challenge with multiple decision variables. A total of 1059 commercially available power devices spanning various voltage, current, and technology categories (e.g., Si, SiC, and GaN) were initially compiled and evaluated against three key criteria.

- 1) *Small Footprint*: As shown in Fig. 3, the form factor and available surface area impose strict constraints. Based on (1), the available area per inverter module is 6875 mm², as dictated by the motor's integrated geometry. This area must accommodate the power devices, gate driver circuitry, terminals, decoupling capacitors, and connectors. To ensure design feasibility, 50% of this area (3437.5 mm²) was allocated to power device placement, eliminating any device exceeding this footprint. It is important to note that while the inverter module can be extended vertically (in the z -direction) due to the absence of volume constraints, maintaining all the components of the FB converter on a single PCB is strongly preferred to minimize complexity and improve system reliability.
- 2) *Weight*: To meet the gravimetric power density target of 45 kW/kg, lightweight components were prioritized during the selection process. This focus ensures compliance with aviation-specific constraints, where weight is a critical performance factor.
- 3) *Efficiency*: A high efficiency target of 98% or greater necessitated devices with minimal conduction and switching losses. The following figure of merit (FOM) was used to evaluate candidate devices:

$$\text{FOM} = R_{\text{ds,on}} \times Q_g \quad (4)$$

where $R_{\text{ds,on}}$ is the on-state resistance and Q_g is the total gate charge of the device. Lower FOM values correspond to lower conduction loss, as demonstrated in [9]. For switching loss considerations, the reverse recovery charge (Q_{rr}) was also minimized wherever applicable.

Using these criteria, all 1059 devices were systematically evaluated, as shown in Fig. 8. The unit weight and footprint of each device are an estimate based on its standard package dimensions. The initial number of parallel devices required to deliver the rated peak load current, is calculated as follows:

$$N_{\text{par,init}} = \text{ceil}\left(\frac{I_{\text{coil,rms}} \times \sqrt{2}}{i_{\text{rated,switch}}}\right) = \text{ceil}\left(\frac{379 \text{ A}}{i_{\text{rated,switch}}}\right) \quad (5)$$

where $i_{\text{rated,switch}}$ is the continuous drain current of the device at 25 °C, as specified in the manufacturer's datasheet. It is important to emphasize that the value of $N_{\text{par,init}}$ in (5) represents the initial estimate of the minimum number of parallel devices per switching position. This estimation is

TABLE II
TOP TEN CANDIDATES FOR THE POWER DEVICES

Part Number	Technology	Voltage (V)	Current (A)	$R_{ds,on}$ (m Ω)	Q_g (nC)	FOM (p Ω C)	$N_{par,init}$	Total Surface (mm ²)	Total Weight (g)
AUIRF7799L2TR	Si	250	375	38	165	6270	1	259.8	0.8
FF06010FA	SiC	650	140	18	315	5670	3	360	6
UJ4SC075005L8S	SiC	750	120	7.2	164	1180.8	4	480	8
EPC2304ENGRT	GaN	200	102	3.1	24	74.4	4	240	6.4
FF06020FA	SiC	650	101	28	194	5432	4	480	8
MCTL80N20Y-TP	Si	200	80	13	144	1872	5	600	10
C3M0025065L-TR	SiC	650	77	34	111	3774	5	600	10
IPT60R028G7XTMA1	Si	600	75	28	123	3444	5	600	10
FF06030FA	SiC	650	74	45	125	5625	5	600	10
NTBL045N065SC1	SiC	650	73	50	105	5250	6	720	12

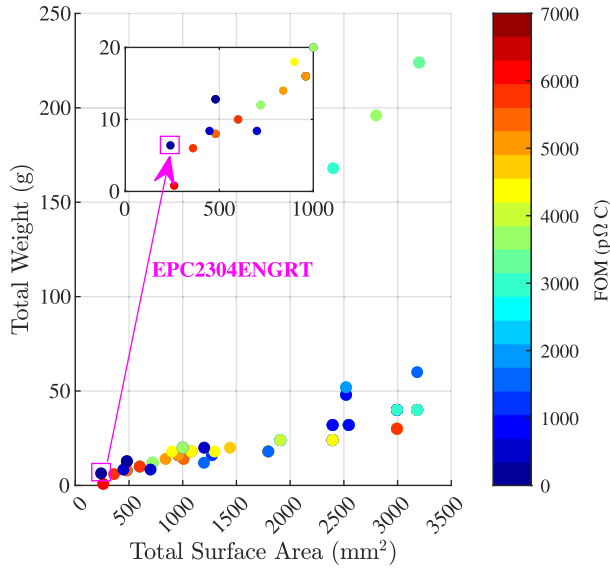


Fig. 8. Switch selection swarm results: The x - and y -axis represent the total surface area and total weight of the devices, respectively, while the color of each point corresponds to the FOM, as indicated by the color bar. For improved visualization, a magnified view highlights the selected switch and its surroundings. The x -axis is constrained to a maximum total surface area of 3500 mm² so that only switches within the maximum surface area limit (3437.5 mm²) are demonstrated.

based solely on datasheet specifications at 25 °C, a temperature point consistently reported across all 1059 candidate devices. This consistent reference ensures a fair and uniform basis for comparison. However, this initial estimate does not account for thermal performance at elevated operating temperatures. To address this, a more accurate assessment of the required number of parallel devices—reflecting actual operating conditions—will be carried out, for the selected device, through detailed electrothermal simulations in the subsequent section.

The total weight and surface area of each switch were calculated by multiplying its estimated unit weight and surface area, based on its standard packaging, by $N_{par,init}$. Consequently, over 900 power devices were excluded since their surface area exceeded the threshold value of 3437.5 mm². Devices with the minimum FOM, surface area, and weight were prioritized.

Table II lists the top ten candidates that ranked highly in at least two of the three selection criteria. The pre-

sented electrical information in the table for each device is collected from its datasheet while the case temperature is at 25 °C. These candidates include devices from multiple semiconductor technologies (Si, SiC, and GaN) and voltage/current ratings. Among these, the EPC2304ENGRT emerged as the optimal choice, excelling in both FOM and surface area. While AUIRF7799L2TR and FF06010FA are lighter than EPC2304ENGRT, their FOM values are two orders of magnitude higher, indicating significantly greater conduction losses. This aligns with their higher $R_{ds,on}$ compared to that of EPC2304ENGRT. Additionally, since GaN devices have no reverse recovery losses, the switching losses for EPC2304ENGRT are expected to be lower than those of AUIRF7799L2TR and FF06010FA.

Now that the topology and the switches are selected, the next step is to conduct an electrothermal study where the thermal performance of the power converter can be studied.

C. Electrothermal Study

In this section, the thermal performance of the selected switch and topology is simulated under the rated condition. Given that the system consists of 18 identical modules, the performance of a single inverter module can be the representative of the entire drive system (IMMD). Therefore, one inverter module is modeled alongside a motor coil and the cooling system using PLECS software, as illustrated in Fig. 9. The blue parts of the circuit represent the cooling system model, while the rest is the electrical system model. The motor coil is modeled as an RLV load, where the voltage source $E(t)$ represents the back-EMF generated by each coil, the resistor R_{coil} models the coil resistance, and the inductor L_{coil} models the coil inductance. Also, Q_{1-4} are four switching positions in the FB converters.

Since this is an electrothermal simulation, identifying the sources of heat generation is crucial. In power converters, the primary contributors to heat generation are the switching devices. FETs generate heat during conduction (conduction loss) and during the turn-on and turn-off transients (switching losses). The conduction loss (P_{cond}), caused by the on-state resistance of the switch, is calculated using (6), where $I_{sw,rms}^2$ is the square of the switch rms current conducted by the switch. Switching loss depends on the dc-link voltage (V_{dc}), load current (I_{Load}), switching frequency (f_{sw}), voltage rise and fall

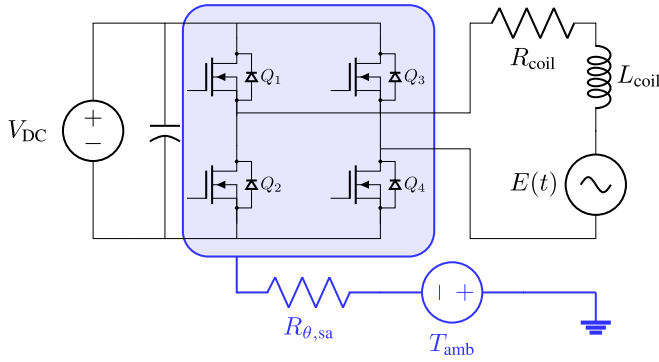


Fig. 9. Electrothermal simulation model including FB topology and motor coil model (RLV).

times (\$t_{rv}\$ and \$t_{fv}\$), and the current rise and fall times (\$t_{ri}\$ and \$t_{fi}\$), which can be calculated through (7) and (8). The total loss generated by each FET is the sum of conduction and switching losses as is shown in (9)

$$P_{cond} = R_{ds,on} \times I_{sw,rms}^2 \quad (6)$$

$$P_{sw,on} = \frac{1}{2} (t_{ri} + t_{fv}) V_{dc} I_{Load} f_{sw} \quad (7)$$

$$P_{sw,off} = \frac{1}{2} (t_{rv} + t_{fi}) V_{dc} I_{Load} f_{sw} \quad (8)$$

$$P_{loss} = P_{cond} + P_{sw,on} + P_{sw,off} \quad (9)$$

In the performed simulations, the switch thermal model is used to calculate all mentioned losses. This model was developed through implementing (6)–(9) in a series of LT-SPICE simulations using data provided by the switch manufacturer.

A clear understanding of the thermal network analogy, derived from the physical structure, is essential for accurately interpreting the electrothermal simulation results. In Fig. 10, the right side shows the physical assembly of the GaN switch and cooling system. A thermal interface material (TIM) is placed between the GaN switch and the heat spreader/heat sink to enhance heat transfer while electrically isolating the switch from the heat spreader/heat sink. The left side shows the corresponding thermal network, with each circuit element colored to match its physical counterpart. The TIM material's thermal resistivity (\$R_{\theta,TIM}\$) and the heat spreader/heat sink to ambient thermal resistivity (\$R_{\theta,Sa}\$) are determined by the designer. This thermal network is included in the performed simulation using PLECS software.

As mentioned earlier, since the rated current of the selected FET is lower than the target load current (268 Arms), it is necessary to parallel GaN FETs. Increasing the number of parallel FETs reduces the current share and per-switch loss, potentially relaxing cooling system requirements. However, it also increases costs, gate drive requirements, and complexity, making it crucial to balance the number of parallel devices and the cooling ability of the TMS.

Key parameters, including the dc-link voltage, load current, and coil specifications, are predetermined. However, variables like the number of parallel devices, switching frequency, and thermal resistance of the heat spreader/heat sink must be determined. An iterative simulation assesses the impact of these parameters on the electrothermal performance of the

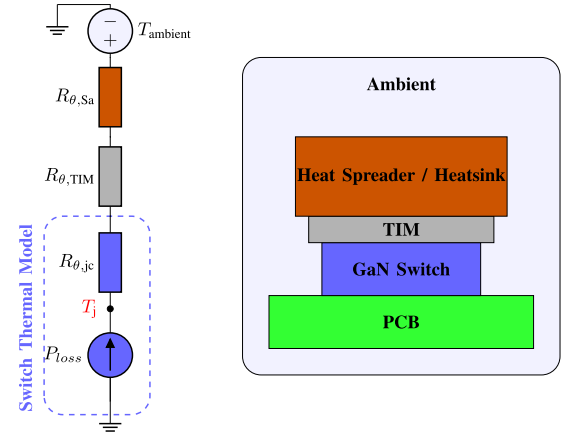


Fig. 10. Thermal network analogy (left) and side view of the physical assembly (right).

inverter module. The number of parallel devices varies from 4 [the initial value obtained by (5)] to 10, switching frequency from 15 to 40 kHz in 5 kHz increments, and the thermal resistivity of the heat spreader/heat sink from 0.1 °C/W to 0.7 °C/W in 0.1 °C/W increments. Junction temperature of the GaNs, converter efficiency, and switching and conduction losses are monitored. The suggested TIM by EPC (TG-A1780 X 0.5) with \$R_{\theta,TIM}\$ of 5.1613 °C/W is used in simulations.

The results of the electrothermal simulations at the rated condition mentioned in Table I are illustrated in Fig. 11. The converter's efficiency is shown in Fig. 11(a) for different values of parallel devices per switching position (\$N_{par}\$), \$f_{sw}\$, and \$R_{\theta,Sa}\$ (each surface represents an \$R_{\theta,Sa}\$ value ascending from top to bottom). The red surface marks the lower efficiency limit of 98% based on Table I. Regardless of \$R_{\theta,Sa}\$, the efficiency with four parallel devices is always below 98%. Fig. 11(a) reveals that with \$N_{par} > 4\$, efficiencies above 98%, which is a design target, are achievable.

Fig. 11(b) illustrates the average junction temperature of the GaN switches for different \$N_{par}\$, \$f_{sw}\$, and \$R_{\theta,Sa}\$ (each surface represents an \$R_{\theta,Sa}\$ value ascending from bottom to top). The red surface shows the upper junction temperature limit which is selected to be 125 °C. With 6 parallel devices, junction temperatures below 125 °C can be achieved, but aggressive cooling is required. With \$N_{par} = 8\$, higher \$R_{\theta,Sa}\$ values allow for relaxed cooling requirements while maintaining junction temperatures below 125 °C.

Increasing \$N_{par}\$ to 10 and beyond can further reduce junction temperatures and increase efficiency with relaxed cooling requirements, but comes with certain drawbacks including increased cost and complexity. Thus, \$N_{par}\$ is selected to be 8, meeting both efficiency and junction temperature targets at certain \$R_{\theta,Sa}\$ values. Fig. 11(a) and (b) shows minimal impact of switching frequency on efficiency and junction temperature due to the selected GaN switches' minimal switching losses (\$\leq 3\%\$ of overall loss). Although higher switching frequencies are desired to fully exploit the potentials of the selected GaN FETs, there are certain concerns regarding the motor coil insulation which limit the switching frequency. Hence,

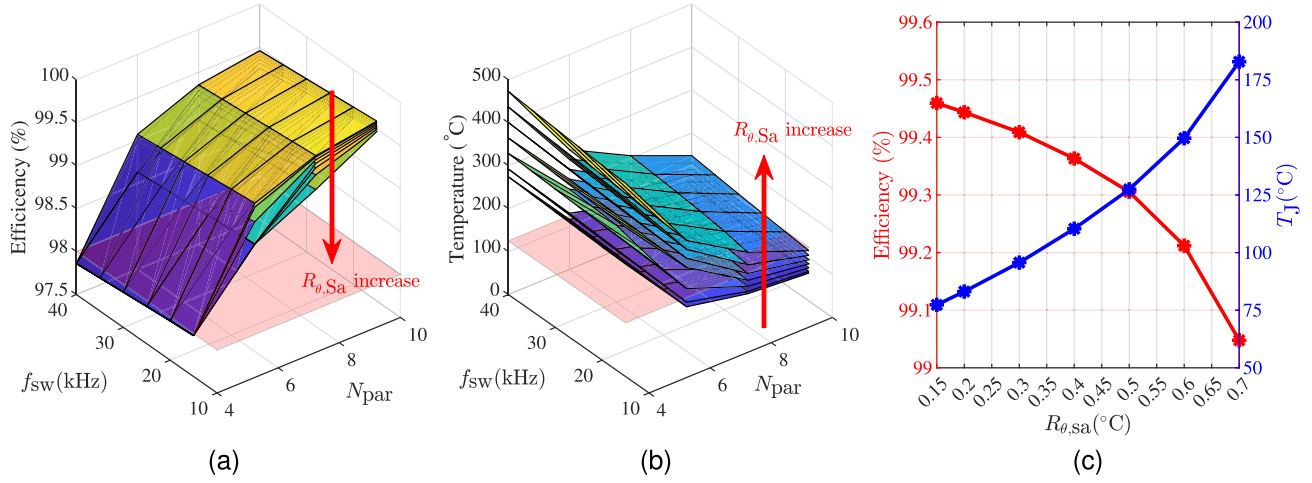


Fig. 11. Electrothermal simulation results. (a) Efficiency of the inverter module for different number of parallel devices at different switching frequencies. (b) Average junction temperature of the GaN switches of the drive unit for different number of parallel devices at different switching frequencies. (c) Efficiency and maximum junction temperature for 8 parallel GaN switches at $f_{sw} = 20$ kHz for different $R_{\theta,Sa}$.

a switching frequency of 20 kHz is selected for the power converter.

While narrowing down the search domain for an optimal design, two of the design constraints are fixed ($N_{par} = 8$ and $f_{sw} = 20$ kHz). Simulation results for varying $R_{\theta,Sa}$ values are shown in Fig. 11(c), with efficiency on the left y-axis and junction temperature on the right y-axis. While the efficiency target is met regardless of $R_{\theta,Sa}$, junction temperature exceeds 125 °C for $R_{\theta,Sa} \geq 0.5$. Therefore, 0.5 °C/W is the maximum allowable $R_{\theta,Sa}$ to satisfy both efficiency and junction temperature constraints.

Fig. 12 demonstrates the overall thermal network for 8 parallel GaNs per switching position which in an FB topology makes it 32 total GaN switches. Based on Fig. 12, the junction temperature for the k th GaN switch can be calculated through the following equation:

$$T_{jk} = (R_{\theta,jc} + R_{\theta,TIM})P_{Sk} + R_{\theta,Sa} \sum_{i=1}^{32} P_{Si} + T_{ambient}. \quad (10)$$

Assuming equal loss generation in all switches, the equivalent thermal resistance between case and the ambient can be calculated as follows:

$$R_{ca} = R_{\theta,TIM} + 32 \times R_{\theta,Sa} \quad (11)$$

where R_{ca} is the equivalent thermal resistivity in steady state from case of a switch to the ambient. Considering the obtained simulation results ($R_{\theta,Sa,max} = 0.5$) and the preset value for R_{TIM} , the maximum value of the case to ambient thermal resistivity per switch is calculated as follows:

$$R_{ca,max} = 5.1613 + 32 \times 0.5 = 21.1613 \text{ } ^\circ\text{C/W}. \quad (12)$$

This value is the starting point of selecting/designing the TMS.

III. PCB DESIGN CONSIDERATIONS

After selecting the topology, the switch, and required number of (parallel) switches, the next step is designing and laying out the PCB. This is a highly complex design challenge due to the high load current, limited space, and the need to

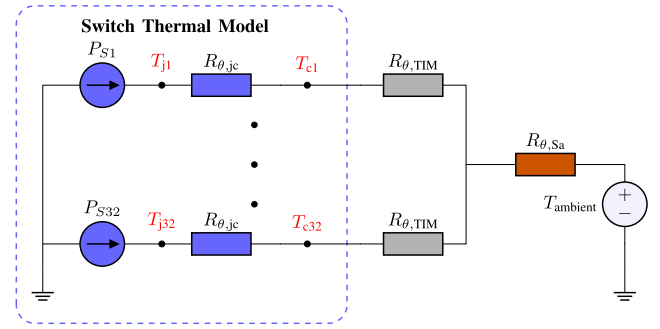


Fig. 12. Loss generation and dissipation thermal network in steady state.

parallel 8 GaN switches. This section will discuss the details of the PCB design and layout, presenting supporting evidence to illustrate the process and considerations involved.

High switching speed and low switching loss are intrinsic properties of WBG devices due to their material physics. Considering the form factor, the loss density of the selected GaN FETs is significantly lower than any silicon counterparts. However, a nonoptimal PCB design can introduce parasitic components that, together with the switching devices, form a resonant circuit. The high switching speed of GaN switches will amplify the effects of this resonance, leading to extreme overvoltages across the GaN switches, potentially resulting in device failure. Therefore, it is crucial to have an optimized PCB design and layout to prevent significant overshoot and voltage resonances.

Another design challenge is the limited space. This is addressed by distributing major functions of the inverter modules across separate PCBs. The function and design of each of these PCBs are explained thoroughly in the following sections.

A. GaN-Based FB Converter PCB

Fig. 13 illustrates the high-frequency (HF) current commutation loop (HF power loop), including an HB circuit with S_1 as the high-side (HS) switch, S_2 as the low-side (LS) switch,

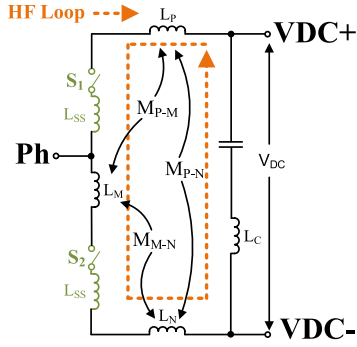


Fig. 13. HF current commutation loop in an HB [4].

HF dc-link capacitor, and parasitic components. To achieve an optimized design, the total loop inductance of the HF loop must be minimized. Using the theory of partial inductance, the loop inductance can be calculated as follows:

$$L_{\text{Loop}} = L_{\text{BB}} + L_C + 2L_{\text{SS}} \quad (13)$$

where L_{Loop} is the loop inductance, L_{BB} is the parasitic inductance of the GaN-based FB converter PCB, L_C is the parasitic inductance of the HF capacitor, and L_{SS} is the parasitic inductance of the GaN switches. While the designer cannot control L_C and L_{SS} , the busbar design directly determines L_{BB} . Therefore, to minimize the loop inductance, L_{BB} must be minimized. Fig. 13 shows self-partial inductances (SPIs) like L_p and mutual partial inductance (MPI) between the interconnected busbar components like M_{p-n} . Proper component layout and configuring the PCB layer stack to benefit from mutual flux cancellation can significantly minimize L_{BB} and eventually L_{Loop} . The power loop inductance was calculated using ANSYS Q3D and the in-depth discussion about that is presented in [4], where L_{Loop} was measured to be 0.7 nH and the parasitic inductance of the GaN-based FB converter PCB (L_{BB}) is calculated to be 0.39 nH.

Considering the high current demand, a 20-layer PCB busbar is designed, with each layer utilizing 2 Oz copper. Fig. 14 demonstrates the PCB layer stack-up, the relative locations of HF power loop components, and two different via types. For optimal layer stack-up, layers VDC+ and VDC- are placed adjacent to each other to achieve maximum mutual flux cancellation, directly helping to reduce L_{BB} . There are two groups of capacitors on the GaN FB converter PCB: group Cap 1, closest to the GaN switches, and group Cap 2, placed beneath the GaN switches on the other side of the board. Cap 1 group consists of the same HF capacitors which are shown in Fig. 13, also known as decoupling capacitors. Decoupling capacitors help stabilize the voltage in the HF power loop during switching transients by providing a low-inductance path for HF current. Cap 2 group also helps stabilize the voltage and can be considered as a local bulk energy source during switching transients since they are also very close to the GaN FETs. Given the high load current demand, local energy storage is crucial to avoid significant voltage fluctuations during switching transients.

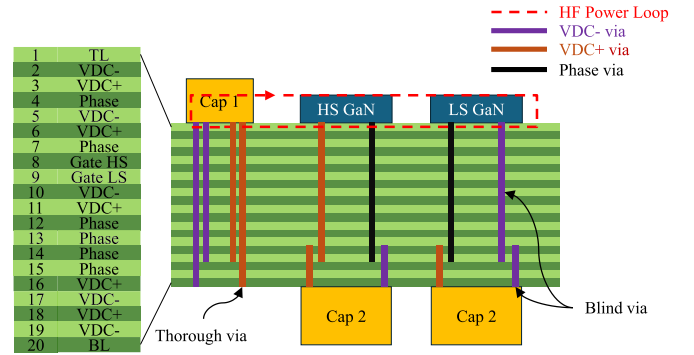


Fig. 14. PCB layer stack-up and via representation.

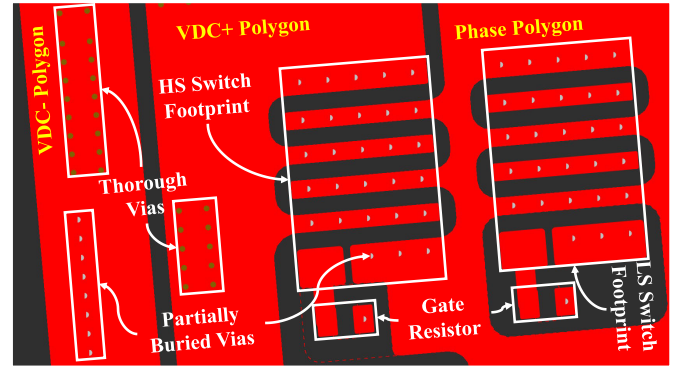


Fig. 15. Details of the top layer of the GaN FB converter PCB.

Two different types of via are shown in Fig. 14. Through vias go through all layers, connecting layers of the same electrical node. However, partially buried vias start from the top/bottom layer and end in one of the internal layers. Due to space constraints, it was impossible to place Cap 2 group beneath the GaN switches using through vias. To address this issue, partially buried vias are used for placing GaN switches and Cap 2 group. From the top side, the partially buried vias of the GaN switches are brought down to layer 17, while the partially buried vias of the Cap 2 group extend from the bottom layer to layer 16. Layers 16 and 17 are VDC- and VDC+, respectively, where partially buried vias from the top and bottom layers meet. This design allows for more local energy storage on the GaN FB converter PCB, improving dc-link voltage stability. Fig. 15 provides a detailed view of the top layer configuration of the GaN-based FB converter PCB. It shows the HS and LS switches, along with the polygon (power plane) layouts, partially buried and through vias, and gate drive resistors. A key challenge in such a compact design is coordinating the polygon layout with clearance rules to prevent internal short circuits within a layer. Additionally, the placement of partially buried vias beneath the GaN FETs and the Cap 1 group is highlighted.

Another significant challenge in this study is ensuring equal current sharing among the 8 paralleled GaN FETs. Furthermore, the distance between the gate driver and the FETs should be minimized to reduce gate loop inductance, another form of parasitic. Therefore, gate signal routing is essential to ensure simultaneous turn-on and turn-off for all parallel FETs.

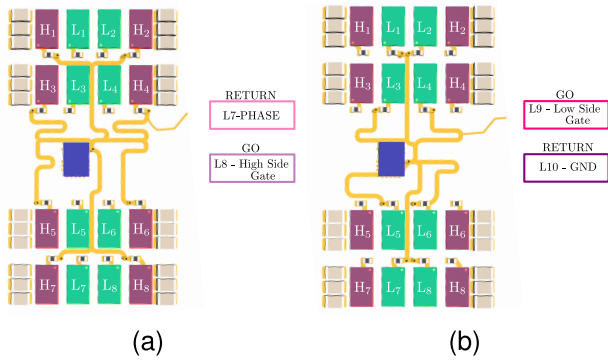


Fig. 16. Layout of the gate drive signals. HS FETs are indicated with purple, while LS FETs are shown in green. LMG1210 gate driver is shown in blue: (a) HS gate signal routing and (b) LS gate signal routing [34].

Given the limited space, the LMG1210 gate driver capable of driving an HB is selected. Ideally, FETs should be arranged in a circular layout around the gate driver for optimal signal distribution. However, due to space constraints, the HB is divided into two switching blocks, each containing 8 GaN FETs. Each block consists of four HS and LS pairs, referred to as power poles. The gate driver is centrally positioned between the two switching blocks to minimize signal path length and ensure efficient operation.

Gate signals for the HS and LS FETs are routed through vias in layers 8 and 9, respectively. Fig. 16 demonstrates the HS and LS gate signal routing, ensuring that all tracks have the same length to maintain symmetry.

Fig. 17 illustrates the layout of the top and bottom layers and also VDC+, VDC-, and phase layer polygon geometry. For example, in the layer stack-up shown in Fig. 14, the layer 4 which is Phase, is laid out as shown in Fig. 17(b). Using polygon instead of traces will reduce the current path inductance and consequently the conduction loss within the PCB. Fig. 18 illustrates the current density distribution across all layers of the PCB, with the results for each layer overlaid. In other words, the current density vectors ($\vec{J}$) for multiple layers were overlaid to observe their cumulative effect. As shown in Fig. 14, multiple layers are allocated to the VDC+, VDC-, and phase nets. This design choice provides parallel paths for the high load current, effectively reducing the overall conduction loss within the PCB. To determine the optimal layer count and copper thickness, a current density analysis was conducted using ANSYS SIwave software. In this finite element analysis (FEA), one specific switching state is shown where the midpoints of the FB topology legs were externally shorted, while the HS switches on the left leg and the LS switches on the right leg were conducting. A current source connected to the dc-link terminals is injecting 192 A to the FB converter. The 192 A current value corresponds to the rms current carried by each switching position in the FB topology under rated load conditions.

As expected, the current density peaks near the conducting HS switches on the left leg and LS switches on the right leg. The FEA results also highlight localized pinch points, such as the regions near the load connection terminals and areas beneath the conducting GaN FETs, where current density

is at its maximum. The current density is highly dependent on the current path geometry which is why the pinch points are observed beneath the switches where their pads surface is extremely small leading to a high current density. Also, the demonstrated green areas on both legs are representing high current density due to the geometry of the polygons in the phase layers as shown in Fig. 17(b), where there is a narrow area. Despite these localized high-current-density regions, the overall current density across the PCB remains below 35 A/mm². This value is manageable with 2 Oz copper per layer, large polygon areas (approximately 5000 mm²), and an effective TMS. Based on the FEA results, the overall calculated conduction loss within the PCB layers is 12.8 W.

Considering all these technical details, the GaN FB converter PCB was designed and manufactured, as shown in Fig. 19. The GaN FB converter PCB in the final application has a semi V-shape design, as sketched on the PCB in Fig. 19, but for ease of heat sink mounting in this study, it is in a square shape.

In Fig. 19(a), the red box on the left highlights an HB including 16 GaN switches (8 HS and 8 LS) and the phase terminal. The yellow box shows the gate driver circuitry, and the green box contains an HS switch, an LS switch, and a few decoupling capacitors. The backside of the GaN FB converter PCB containing Cap 2 group, control and communication board connector, dc terminals, and load terminals are also shown in Fig. 19(b). In the manufacturing process of all the PCBs in this study, an electroless nickel immersion gold (ENIG) finish is used to ensure high quality and facilitate the mounting of SMD components. Additionally, all the vias are capped, plated, and filled with nonconductive material as recommended by the GaN manufacturer to enhance reliability, heat transfer, and performance.

B. DC-Link Capacitor PCB

To further enhance the rms current handling capability of the drive unit and filter HF current ripple, a bulk capacitor bank is essential. This capacitor bank also serves as a local energy storage for the high load current demand, helping to stabilize the dc-link voltage. In other words, it is ideally desired that the dc power supply provides only the dc current while the HF components of the current are managed by the dc-link bulk capacitor. Using the method outlined in [35], the rms current requirement for the capacitor bank in a single-phase FB topology can be calculated as in (14). The fundamental assumption is that the switching frequency is significantly higher than the fundamental frequency

$$i_{c,rms} = \sqrt{m \hat{I}_N^2 \left[\frac{2}{3\pi} + \cos^2(\phi) \left(\frac{2}{3\pi} - \frac{m}{4} \right) \right]} \quad (14)$$

where $i_{c,rms}$ is the rms value of the capacitor current, m is the modulation index, $\hat{I}_N$ is the peak value of the load current, and ϕ is the power angle. Using the value provided in Table I and (14), the rms value of the dc-link capacitor current is calculated as 152.32 A. Given $i_{c,rms}$, the number of required capacitors is calculated and the dc-link capacitor PCB is designed, as shown in Fig. 20. The capacitor board

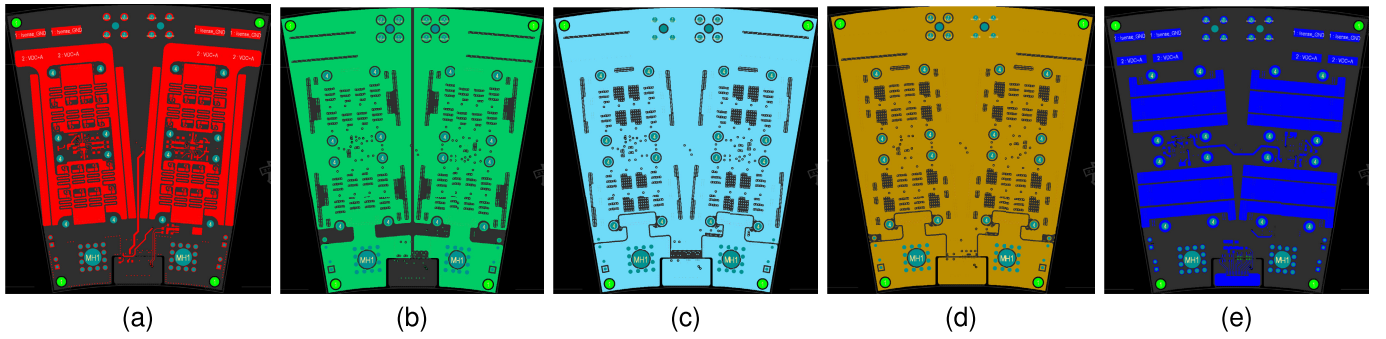


Fig. 17. Polygon layout in different layers. (a) Top layer. (b) Phase layer. (c) VDC+ layer. (d) VDC- layer. (e) Bottom layer.

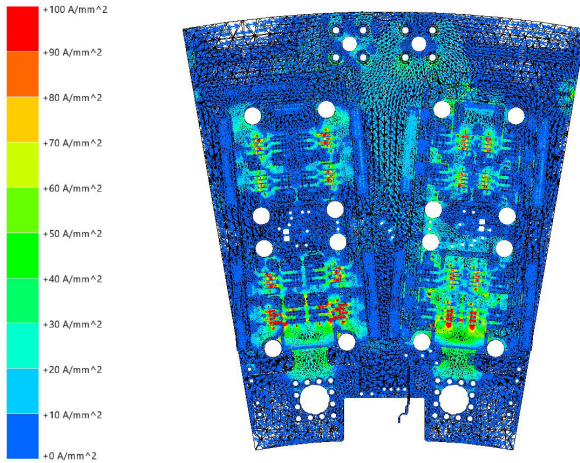


Fig. 18. Current density evaluation of the GaN-based FB converter PCB using ANSYS SIwave during one switching state (HS switches of the left leg and LS switches of the right leg are conducting).

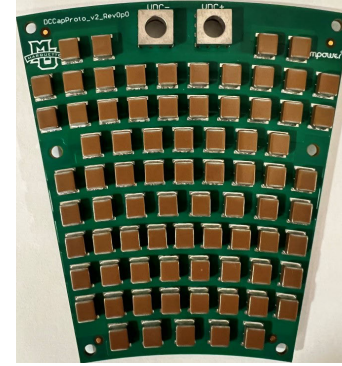


Fig. 20. DC-link capacitor PCB.

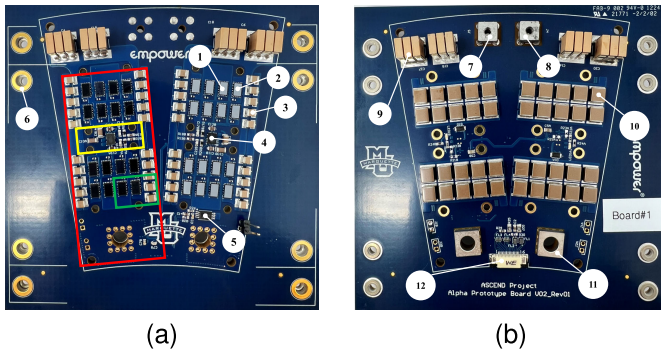


Fig. 19. Manufactured 20-layer GaN FB converter PCB. (a) Front view: 1) LS GaN FET; 2) HS GaN FET; 3) decoupling capacitor (Cap 1); 4) gate driver; 5) ACS37612 phase current sensor; and 6) heat sink mounting hole. Also, the red box demonstrates an HB, the yellow box emboldens the gate driver circuitry, and the green box shows a power pole. (b) Rear view: 7) VDC- connector; 8) VDC+ connector; 9) CeraLink capacitor array; 10) local dc-link ceramic capacitors (Cap 2); 11) phase terminal; and 12) control and communication board connector.

is an 8-layer PCB utilized with 2 Oz copper, and employs 156 ceramic capacitors (CGA9P3X7T2E225K250KA) on both sides, totaling 343.2 μF .

C. Control and Communication PCB

As detailed in Section II, the IMMD system in this study is composed of 18 identical inverter modules, each equipped with

a dedicated control and communication board. This design ensures that every inverter module is a fully autonomous power converter, capable of independent operation. The control and communication board, illustrated in Fig. 21, integrates the TMS320F280049C microcontroller from Texas Instruments' C2000 family.

The TMS320F280049C microcontroller is equipped with a high-speed and robust communication protocol known as the fast serial interface (FSI). This protocol facilitates rapid and reliable communication between the 18 inverter modules and the main control unit, forming the backbone of the fault-tolerant performance of the IMMD. The advanced control and communication platform is instrumental in achieving the high reliability and operational flexibility required by the system.

As shown in Figs. 4 and 7, the overall system topology necessitates two primary control objectives. First, the dc source voltage must be evenly distributed among the six series-connected three-phase subsets, which requires the implementation of a dc voltage balancing control loop. Second, the load torque demand must be tracked accurately, necessitating the implementation of a current control loop [36]. Achieving these control objectives is contingent on high-speed and reliable communication between the inverter modules, which was a pivotal factor in selecting the TMS320F280049C microcontroller with its FSI capabilities.

Additionally, the control board houses the gate driver's power supply and the signal conditioning circuitry for

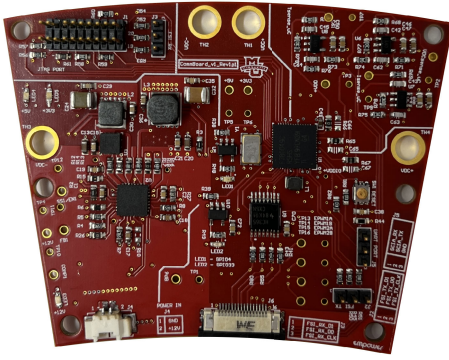


Fig. 21. Communication and control PCB.

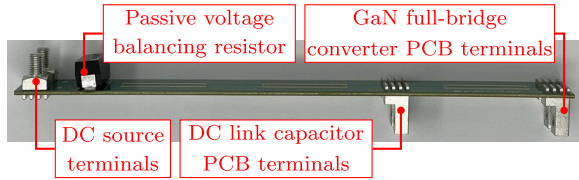


Fig. 22. DC-link PCB busbar.

processing the measured variables, further enhancing its integration and functionality.

D. DC-Link Busbar and Full Stack-up

After designing the three main parts of the drive unit, the final step is to connect them effectively with low path inductance. For this purpose, a PCB busbar was designed to link the GaN FB converter PCB to the dc capacitor PCB and provide terminals for the dc power source, as shown in Fig. 22. Optimizing the PCB design to achieve low path inductance is crucial to prevent resonance between the dc-link capacitor PCB and the GaN FB converter PCB during switching transients. The designed PCB busbar is an 8-layer PCB with 2 Oz copper, where the VDC+ and VDC- layers are alternated to maximize mutual flux cancellation.

Notably, the communication board is not connected to the PCB busbar; instead, it is mounted on the bulk dc capacitor board using standoffs. The communication board is connected to the GaN FB converter PCB through a ribbon cable, as shown in the full stack-up of a single inverter module in Fig. 23. The PCB busbar includes footprints for additional dc-link capacitor arrays if more capacitors are needed for dc-link voltage stabilizing during testing.

Fig. 24 presents the mass breakdown of the components in a single inverter module. The PCB accounts for the largest portion, contributing 32% of the total mass, while the GaN power devices contribute only 5%. Additionally, the designed inverter module achieves an overall power density of 43.96 kW/kg (TMS excluded), which is slightly below the design target outlined in Table I.

At this stage, the design phase of the IMMD is concluded with a fully designed and stacked single inverter module. All other modules will be exact replicas of the one shown in Fig. 23. The next step involves testing the design's perfor-

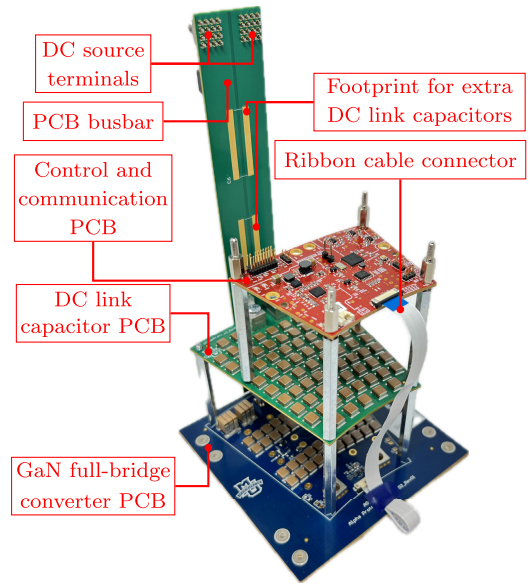


Fig. 23. Full stack-up of a single inverter module including the GaN FB converter PCB, dc-link capacitor PCB, communication PCB, and the dc busbar PCB.

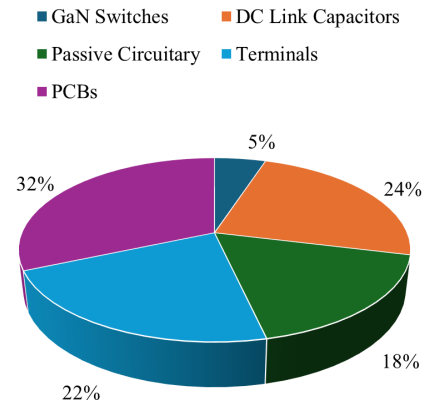


Fig. 24. Mass breakdown of the components of an inverter module. The total inverter module weight is 316.94 g.

mance. In the subsequent sections, various test results will be presented to evaluate the design's effectiveness.

IV. SWITCHING CHARACTERISTICS

Given the utilization of GaN FETs in this design, the first step is to evaluate the performance during switching transients. To achieve this, DPTs were conducted at different dc-link voltages and load currents on the test circuit shown in Fig. 25. Due to the symmetrical design, the results are similar for both HB legs, so only a single leg is tested. In the DPT setup, an inductor with the value of $L_{DPT} = 2.47\mu\text{F}$ is connected across the LS switch, while the drain-source voltage of the LS switch and the dc-link voltage are monitored to investigate the turn-off voltage overshoot. The waveforms are captured using a Tektronix MSO58 oscilloscope with a 6.25 GS/s sampling rate, using a Tektronix TPP1000, 1 GHz single-ended probe for LS switch drain-source voltage and dc-link measurement, and a PEM CWT Rogowski current waveform transducer for

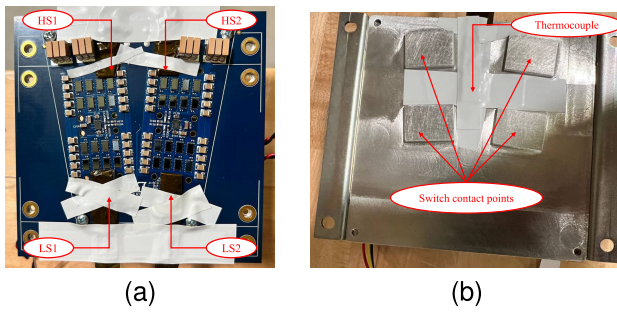


Fig. 28. Thermocouple locations (a) on the GaN FB converter PCB close to HS and LS FETs and (b) on the heat sink in the middle.

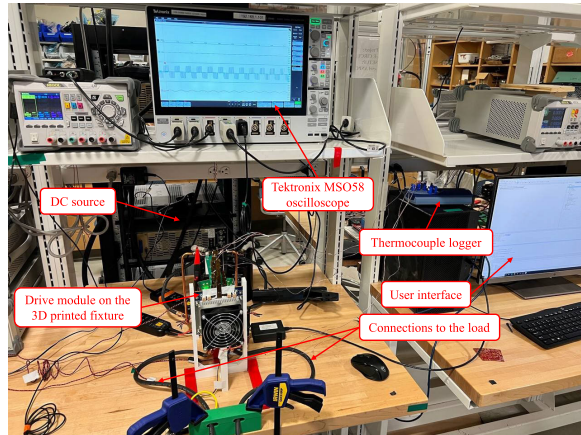


Fig. 29. Test setup for continuous ac testing.

module, providing adequate space for both load and dc source connections, as well as air ventilation for the heat sink fan.

Although the final application incorporates a heat pipe-based cooling system, the experimental setup utilized an off-the-shelf heat sink and fan solution (FH 9040MU) with a thermal resistance of $0.15\text{ }^{\circ}\text{C/W}$, as specified in its datasheet. The thermal resistance value of the selected heat sink is lower than $0.5\text{ }^{\circ}\text{C/W}$, which is the $R_{\theta, \text{Sa, max}}$ that was determined by the electrothermal simulation. The loss and temperature data collected during the experimental testing will serve as a reference for the design and evaluation of the heat pipe-based TMS. Specifically, the thermal resistance of the off-the-shelf heat sink and fan TMS provides a basis for comparison with the thermal resistance of the designed heat pipe-based TMS. Additionally, the recorded temperatures will be used as a benchmark to validate and optimize the performance of the heat pipe-based TMS.

To monitor the temperatures of the GaN FETs, four Type T thermocouples were strategically positioned near the HS and LS FETs. Additionally, one thermocouple was placed on the heat sink, as shown in Fig. 28. While these thermocouples are located in close proximity to the GaN FET cases, it is important to note that they do not directly represent the case temperature or the junction temperature. However, their minimal distance from the case provides a reasonable estimate of the GaN FET temperatures, offering valuable insight into the thermal behavior of the inverter module. The test setup for the ac testing of the module is shown in Fig. 29. During the

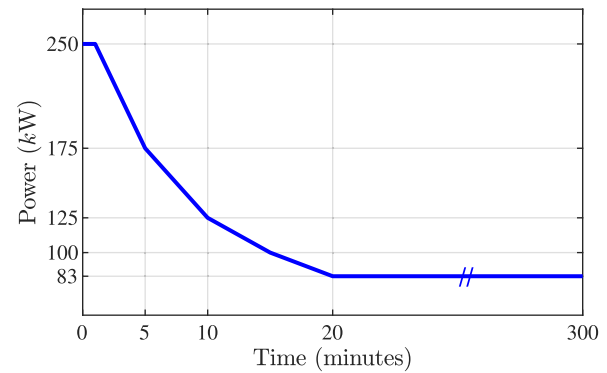


Fig. 30. Mission profile [40].

ac test, the module was connected to the dc source and the hand-wound inductors to simulate the motor coil's behavior.

Fig. 30 illustrates the mission profile for the target application, which is an all-electric aircraft propulsion system. During the takeoff phase (first 2 min), the system operates at rated power, representing the highest power demand. Following this phase, the power decreases progressively as the aircraft transitions to the cruise condition, where the system operates at approximately one-third of the rated power.

To validate the designed inverter module, tests were conducted at multiple operating points ranging from 20% of the load current to the full load current. Among these, the rated power condition represents the worst case scenario in terms of thermal loading, as it generates the maximum loss within the inverter module. The primary objective of the ac tests was to evaluate the module's performance under varying load currents while closely monitoring the temperature rise. This ensures that the design is thermally robust and capable of sustaining the operational requirements of the mission profile.

Temperature data was recorded using a Pico Technology USB TC-08 thermocouple data logger, which sampled the thermocouples' readings every 500 ms and applied averaging method for accuracy. The dc-link voltage and load voltages were measured using TMDP0200 differential probes. The current drawn from the dc source was measured using a TCP0150 current probe, while the load current was measured with a CWTMini 3B Rogowski coil.

Using the Code Composer Studio IDE, the user can command and control the microcontroller, inputting the modulation index, switching frequency, fundamental frequency, and all other necessary settings. During the test procedure, the modulation index was gradually increased from 0.05 to 0.247. After each increment, there was a waiting period to allow the system to reach thermal steady state before recording the results. In the final stage, with a modulation index of 0.247 and a current of 268 Arms, the test was conducted for 16 min.

Fig. 31 illustrates the test results at 166 VDC, where the inverter module delivered 268 Arms to the load. A zoomed-in per unit illustration of the last four fundamental cycles is also shown in Fig. 32. During this high-current test, the average temperature recorded by the four thermocouples placed near the GaN FETs on the FB converter PCB was $77.51\text{ }^{\circ}\text{C}$.

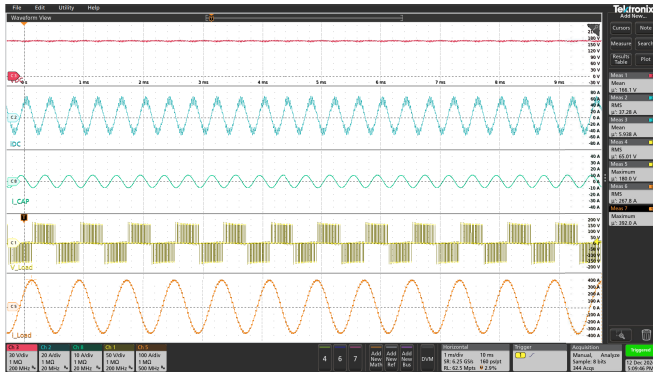


Fig. 31. Experimental test results: $V_{dc} = 166.67$ V, $f_{sw} = 20$ kHz, $f_{fundamental} = 1250$ Hz, $I_{load,rms} = 268$ A. Plots from top to bottom: dc-link voltage, dc source current, load voltage, and load current.

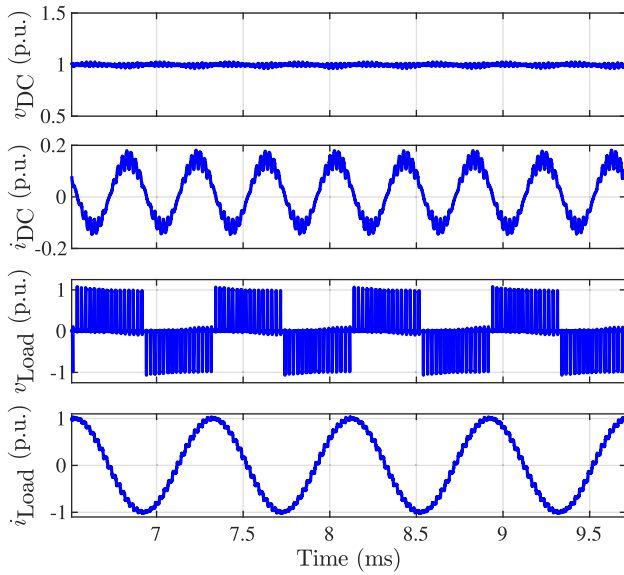


Fig. 32. Zoomed-in view of the four cycles of the scope window in Fig. 31. The base voltage and current for per unit are as follows: $V_{base} = 166.67$ V and $I_{base} = 268$ A.

Additionally, using an FLIR thermal camera, the maximum temperature on the GaN-based FB converter PCB was identified at the load terminal, measuring 104.3 °C. The recorded temperatures remained below the GaN FETs' upper design limit of 125 °C and the PCB's temperature threshold. Hence, the results confirm that the module successfully delivered the rated power while maintaining thermal stability, thereby validating the effectiveness of the design.

To evaluate the efficiency of the designed inverter module under rated load conditions, both input and output power must be accurately measured. To ensure high accuracy and eliminate phase delay errors, a high-precision power analyzer (PPA5500) was used for all power measurements. Unlike other measurement methods that may introduce signal delays, the power analyzer provides reliable and synchronized measurements. This is particularly important in this setup, where the inverter is supplying a highly inductive load and even small phase errors can significantly affect power calculations. Fig. 33 illustrates the generated losses inside the inverter module and

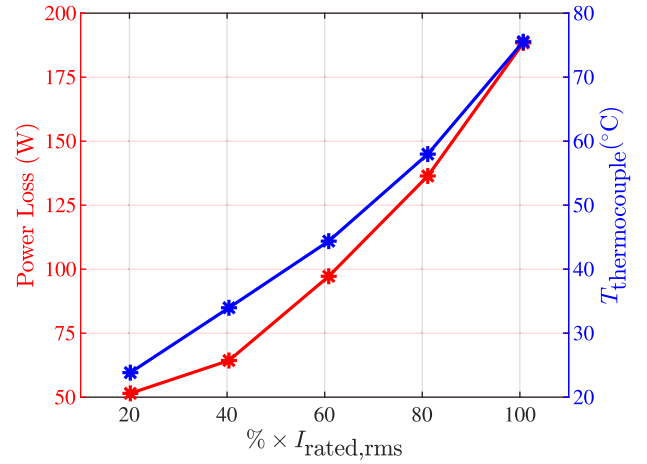


Fig. 33. Generated loss in an inverter module and the recorded temperature by thermocouples during different loading.

the average temperatures recorded by the four thermocouples placed on the GaN-based FB converter PCB that are shown in Fig. 28(a). At the rated current, where the inverter module is delivering its rated power, the total generated loss within the inverter module is calculated to be 189.25 W. This value accounts for not only the losses in the GaN FETs but also all other passive losses within the inverter module. Considering that the rated active power for each inverter module is $1/18$ th of the overall system power (250 kW), the efficiency of the designed inverter module is calculated to be 98.65% . The obtained efficiency meets the design target of 98% , validating both the design process and its effectiveness.

While the reported power density is slightly below the initial target, it is important to note that minor design refinements during the integration phase—when the inverter module is mechanically coupled with the motor and cooling system—can lead to further improvements. For example, reducing the length of the dc-link busbar PCB can decrease the module's overall weight, thereby enhancing power density. Additionally, optimizing connection points and terminal designs can minimize power losses and potentially increase efficiency. These opportunities for refinement highlight key areas for future optimization to further elevate system performance.

VI. CONCLUSION

In this article, the design and experimental validation of a single inverter module for a 250 kW IMMD were presented. The module is a fully autonomous power converter consisting of a GaN-based FB converter PCB, a dc-link capacitor PCB, a control and communication PCB, and an interconnecting PCB busbar. This modular design enables each unit to operate independently while integrating seamlessly into the IMMD structure, facilitating communication and coordination with other modules as needed.

The rigorous selection process for the optimal power switch, involving 1059 initial candidates and evaluated based on surface area, weight, and a defined FOM, was detailed. The top ten candidates were presented, and the EPC2304ENGRT

GaN FET was selected. Using its thermal model, electrothermal simulations determined the number of required parallel switches and the maximum permissible thermal resistance for the TMS. A step-by-step solution to a complex PCB design challenge was provided, targeting optimization of parasitic inductance in the power loop and maintaining acceptable current density in the PCB layers. The resulting inverter module achieved a specific power of 43.96 kW/kg.

DPT at rated load current demonstrated a voltage overshoot of less than 4%, confirming the effectiveness of the GaN FB converter PCB design in minimizing loop inductance. The recorded turn-on and turn-off dv/dt values were 8.16 and 8.42 V/ns, respectively—both within the recommended range of 2–10 V/ns for motor drive applications.

The module was further tested under rated conditions (166.67 VDC and 268 Arms), where the average temperature near the GaN FETs was 77.51 °C, well within the junction temperature design limit of 125 °C. During the experimental testing at rated condition, the module achieved an efficiency of 98.65%.

While the reported efficiency meets the design target, the inverter module's power density falls slightly short. This minor discrepancy can be reduced—or even eliminated—through targeted design improvements, such as shortening the dc-link busbar PCB, which would reduce the overall module mass and increase power density. Additionally, optimizing the load connection points and terminal design can help minimize losses and further enhance efficiency. Although the control strategy, EMI performance evaluation, and the heat pipe-based TMS were not fully explored in this manuscript due to space limitations, these aspects will be addressed in future work.

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